

CivicOne — Digital Identity Platform

Research Foundations, Architectural Paradigms & Scientific References

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1. Executive Summary & Research Vision

CivicOne (CIVIQONE) addresses critical vulnerabilities in legacy centralized identity systems, including identity theft, data breaches, uncoordinated departmental verifications, and unauthorized user tracking. Grounded in *Self-Sovereign Identity (SSI)* and *Aadhaar Data Vault (ADV)* tokenization standards, CivicOne provides a unified digital identity ecosystem where citizens retain cryptographically verifiable control over their personal data credentials.

2. Core Research Paradigms & Architecture

A. Self-Sovereign Identity (SSI) Framework

Traditional identity architectures rely on central identity providers (IdPs) that act as single points of failure and tracking hubs. CivicOne implements W3C-compliant decentralized identity principles, empowering citizens to store credentials locally in an encrypted **CivicVault** and issue scoped, time-bound consent tokens to third-party organizations.

B. Aadhaar Data Vault (ADV) Tokenization

In accordance with UIDAI security mandates, storing raw 12-digit Aadhaar Unique Identification (UID) numbers in third-party relational databases is prohibited. CivicOne implements an ADV Engine that tokenizes Aadhaar numbers into 72-byte **Anonymous User Tokens (AUT)**. All external verification panels operate solely on masked identity references (XXXX XXXX 1001) and AUT tokens.

C. Zero-Knowledge Proofs (ZKP) & Selective Disclosure

To minimize data exposure during verification (e.g., hotel check-ins, age-gate checks, academic verifications), CivicOne provides zero-knowledge selective attribute disclosure. Requesting entities receive cryptographically signed verification proofs (e.g., `IsOver18: true` or `VerificationStatus: VERIFIED`) without receiving full raw PII or physical document scans.

D. Hardened Cryptographic Operations

The security model enforces strict key derivation and constant-time execution standards:

- **PBKDF2 Key Derivation:** HMAC-SHA256 with 100,000 iterations and 16-byte random salts for citizen MPIN hashing (NIST SP 800-132).
- **Constant-Time Comparison:** Verification uses `hmac.compare_digest` to prevent side-channel timing attack vectors.
- **Stateless JWT Bearer Tokens:** Claims are signed with HMAC-SHA256 (RFC 7519) with strict role-based access control (RBAC).

3. Applied Research Workflow Example

The diagram below illustrates the end-to-end verification sequence during an identity authorization request (e.g., organization verification or police clearance):

Step	Actor / Layer	Action / Cryptographic Operation
1	Organization Portal	Issues verification challenge request with required scope.
2	Citizen (React SPA)	Reviews scope, approves consent, enters 4-digit MPIN.
3	Python Auth Engine	Derives PBKDF2 hash, checks constant-time digest, issues 180s challenge.
4	Aadhaar Data Vault	Maps UID to 72-byte Anonymous User Token (AUT) & masked identity.
5	CivicOne Gateway	Returns signed JWT verification proof token. Zero raw PII exposed.

4. Standards & Bibliography References

- **W3C Verifiable Credentials Data Model v1.1 / v2.0**

World Wide Web Consortium (W3C) Recommendation, 2022. Standard for verifiable, tamper-evident digital identity credentials.

- **UIDAI Aadhaar Data Vault Guidelines**

Unique Identification Authority of India (UIDAI), Circular F. No. 11020/219/2017-UIDAI. Standard for UID tokenization and key protection.

- **NIST SP 800-132 (PBKDF2 Specification)**

National Institute of Standards and Technology (NIST), 2010. Recommendation for Password-Based Key Derivation Part 1: Storage Applications.

- **RFC 7519: JSON Web Token (JWT)**

Internet Engineering Task Force (IETF), 2015. Standardized compact, URL-safe representation of claims transferred between two parties.

- **Self-Sovereign Identity (SSI) Architecture**

Allen, C. (2016). 'The Path to Self-Sovereign Identity', Life With Alacrity. Foundational paper on user-controlled identity systems.

- **Bcrypt Key Derivation & Hashing**

Provos, N., & Mazières, D. (1999). 'A Future-Adaptable Password Scheme', USENIX Security Symposium. Adaptive hash function standard.

- **Zero-Knowledge Credentials (ZKP)**

Camenisch, J., & Lysyanskaya, A. (2001). 'An Efficient System for Non-transferable Anonymous Credentials'. Advances in Cryptology — EUROCRYPT 2001.

- **Multilingual Conversational Agents in HCI**

Luger, E., & Sellen, A. (2016). 'Like Having a Really Bad PA: The Gulf Between User Expectation and Experience of Conversational Agents'. ACM CHI Conference.

5. Formatted BibTeX Citations

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  author     = {W3C Working Group},  
  year       = {2022},  
  url        = {https://www.w3.org/TR/vc-data-model/},  
  organization = {World Wide Web Consortium}  
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@techreport{uidai_adv_2017,  
  title      = {Circular on Implementation of Aadhaar Data Vault},  
  author     = {UIDAI},  
  year       = {2017},  
  institution = {Unique Identification Authority of India, MeitY},  
  number     = {F. No. 11020/219/2017-UIDAI}  
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@techreport{nist_sp800_132,  
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  author     = {Tur an, Meltem S"onmez and Barker, Elaine and Kelsey, John},  
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  institution = {National Institute of Standards and Technology},  
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